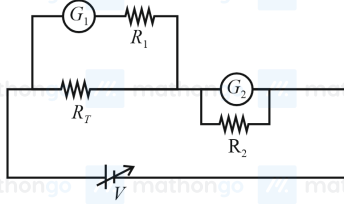


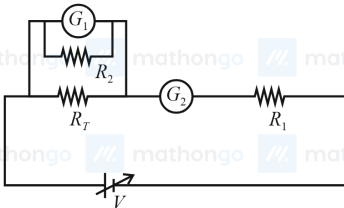
1. A student is provided with a variable voltage source V , a test resistor $R_T = 10\Omega$, two identical galvanometers G_1 and G_2 and two additional resistors, $R_1 = 10M\Omega$ and $R_2 = 0.001\Omega$. For conducting an experiment to verify ohm's law, the most suitable circuit is:

[2023 (06 Apr Shift 2)]

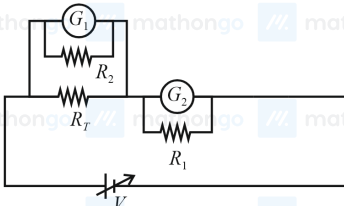
(1)



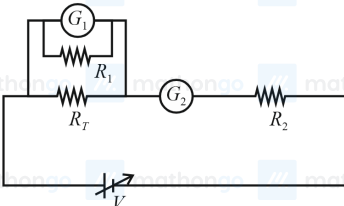
(2)



(3)



(4)



2. Figure shows a part of an electric circuit. The potentials at points a , b and c are 30 V , 12 V and 2 V respectively. The current through the 20Ω resistor will be,

(1) 1.0 A

(2) 0.4 A

(3) 0.6 A

(4) 0.2 A

[2023 (06 Apr Shift 2)]

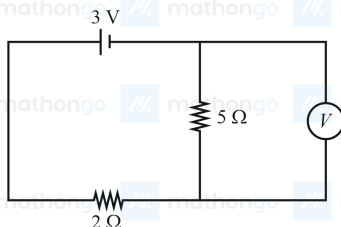
(1) 1.0 A

(2) 0.4 A

(3) 0.6 A

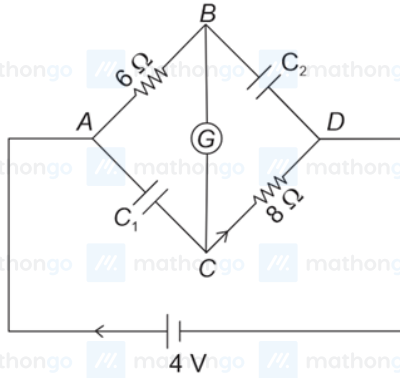
(4) 0.2 A

3. As shown in the figure the voltmeter reads 2 V across 5Ω resistor. The resistance of the voltmeter is _____ Ω



[2023 (06 Apr Shift 2)]

4. In this figure the resistance of the coil of galvanometer G is $2\ \Omega$. The emf of the cell is 4 V . The ratio of potential difference across C_1 and C_2 is



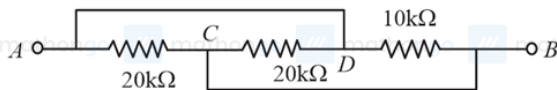
[2023 (08 Apr Shift 1)]

- (1) 1
- (2) $\frac{4}{5}$
- (3) $\frac{5}{4}$
- (4) $\frac{3}{4}$

5. A current of 2 A flows through a wire of cross-sectional area 25.0 mm^2 . The number of free electrons in a cubic meter are 2.0×10^{28} . The drift velocity of the electrons is $\times 10^{-6}\text{ ms}^{-1}$
(given, charge on electron = $1.6 \times 10^{-19}\text{ C}$).

[2023 (08 Apr Shift 1)]

6. The equivalent resistance between A and B as shown in figure is:



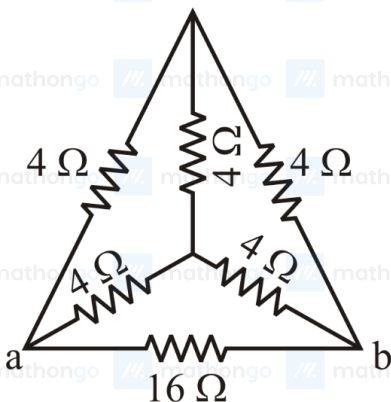
[2023 (08 Apr Shift 2)]

- (1) $10\text{ k}\Omega$
- (2) $5\text{ k}\Omega$
- (3) $20\text{ k}\Omega$
- (4) $30\text{ k}\Omega$

7. The number density of free electrons in copper is nearly $8 \times 10^{28}\text{ m}^{-3}$. A copper wire has its area of cross-section = $2 \times 10^{-6}\text{ m}^2$ and is carrying a current of 3.2 A . The drift speed of the electrons is $\times 10^{-6}\text{ m s}^{-1}$.

[2023 (08 Apr Shift 2)]

8. The equivalent resistance of the circuit shown below between points a and b is :



[2023 (10 Apr Shift 1)]

- (1) $16\ \Omega$
- (2) $3.2\ \Omega$
- (3) $24\ \Omega$
- (4) $20\ \Omega$

9. 10 resistors each of resistance $10\ \Omega$ can be connected in such as to get maximum and minimum equivalent resistance. The ratio of maximum and minimum equivalent resistance will be _____.

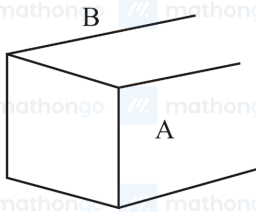
[2023 (10 Apr Shift 1)]

10. In a metallic conductor, under the effect of applied electric field, the free electrons of the conductor

[2023 (10 Apr Shift 2)]

- (1) Drift from higher potential to lower potential
- (2) Move with the uniform velocity throughout from lower potential to higher potential
- (3) Move in the straight line paths in the same direction
- (4) Move in the curved paths from lower potential to higher potential

11. A rectangular parallelepiped is measured as $1\text{ cm} \times 1\text{ cm} \times 100\text{ cm}$. If its specific resistance is $3 \times 10^{-7}\ \Omega\text{ m}$, then the resistance between its two opposite rectangular faces will be $_____ \times 10^{-7}\ \Omega$.



[2023 (10 Apr Shift 2)]

12. Two identical heater filaments are connected first in parallel and then in series. At the same applied voltage, the ratio of heat produced in same time for parallel to series will be:

[2023 (11 Apr Shift 1)]

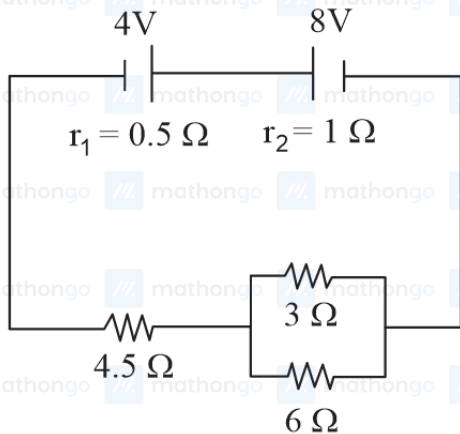
- (1) 1 : 4
- (2) 4 : 1
- (3) 2 : 1
- (4) 1 : 2

13. The current sensitivity of moving coil galvanometer is increased by 25%. This increase is achieved only changing in the number of turns of coils and area of cross section of the wire while keeping the resistance of galvanometer coil constant. The percentage change in the voltage sensitivity will be:

[2023 (11 Apr Shift 1)]

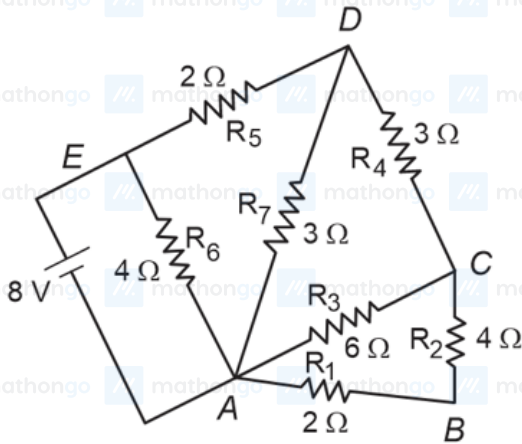
- (1) +25%
- (2) -50%
- (3) -25%
- (4) Zero

14. In the circuit diagram shown in figure given below, the current flowing through resistance $3\ \Omega$ is $\frac{x}{3}\text{ A}$. The value of x is _____.



[2023 (11 Apr Shift 1)]

15.



The current flowing through R_2 is :

[2023 (11 Apr Shift 2)]

- (1) $\frac{2}{3}$ A
- (2) $\frac{1}{2}$ A
- (3) $\frac{1}{3}$ A
- (4) $\frac{1}{4}$ A

16. Two identical cells each of emf 1.5 V are connected in series across a $10\ \Omega$ resistance. An ideal voltmeter connected across $10\ \Omega$ resistance reads 1.5 V. The internal resistance of each cell is $\underline{\hspace{1cm}}$ Ω .

[2023 (11 Apr Shift 2)]

17. A wire of resistance $160\ \Omega$ is melted and drawn in a wire of one-fourth of its length. The new resistance of the wire will be

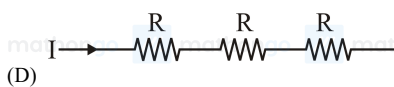
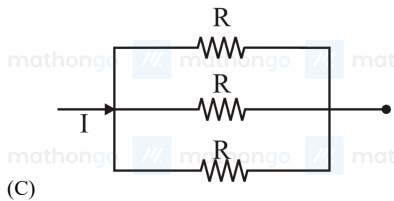
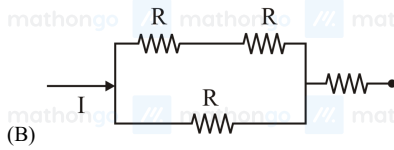
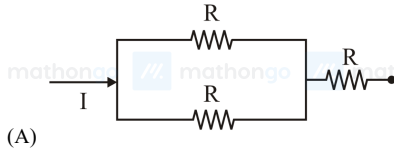
[2023 (12 Apr Shift 1)]

- (1) $16\ \Omega$
- (2) $10\ \Omega$
- (3) $640\ \Omega$
- (4) $40\ \Omega$

18. The current flowing through a conductor connected across a source is 2 A and 1.2 A at 0°C and 100°C respectively. The current flowing through the conductor at 50°C will be $\underline{\hspace{1cm}} \times 10^2$ mA.

[2023 (12 Apr Shift 1)]

19. Different combination of 3 resistors of equal resistance R are shown in the figures. The increasing order for power dissipation is:



[2023 (13 Apr Shift 1)]

- (1) $P_B < P_C < P_D < P_A$
- (2) $P_C < P_D < P_A < P_B$
- (3) $P_C < P_B < P_A < P_D$
- (4) $P_A < P_B < P_C < P_D$

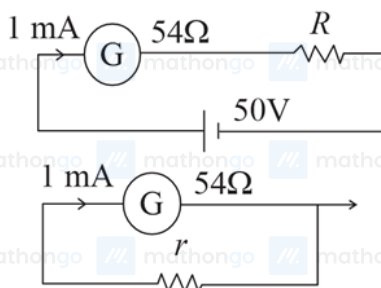
20. A potential V_0 is applied across a uniform wire of resistance R . The power dissipation is P_1 . The wire is then cut into two equal halves and a potential of V_0 is applied across the length of each half. The total power dissipation across two wires is P_2 . The ratio of $P_2 : P_1$ is $\sqrt{x} : 1$. The value of x is _____.

[2023 (13 Apr Shift 1)]

21. When a resistance of $5\ \Omega$ is shunted with a moving coil galvanometer, it shows a full scale deflection for a current of $250\ \text{mA}$, however when $1050\ \Omega$ resistance is connected with it in series, it gives full scale deflection for $25\ \text{V}$. The resistance of galvanometer is _____ Ω .

[2023 (13 Apr Shift 1)]

22. For designing a voltmeter of range $50\ \text{V}$ and an ammeter of range $10\ \text{mA}$ using a galvanometer which has a coil of resistance $54\ \Omega$ showing a full scale deflection for $1\ \text{mA}$ as in figure.



- (A) for voltmeter $R \approx 50\ \text{k}\Omega$
- (B) for ammeter $r \approx 0.2\ \Omega$
- (C) for ammeter $r \approx 6\ \Omega$
- (D) for voltmeter $R \approx 5\ \text{k}\Omega$
- (E) for voltmeter $R \approx 500\ \Omega$

Choose the correct answer from the options given below:

[2023 (15 Apr Shift 1)]

- (1) (A) and (C)
- (2) (C) and (E)
- (3) (C) and (D)
- (4) (A) and (B)

23. Given below are two statements:

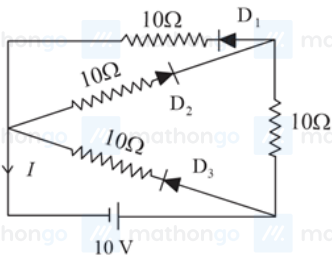
Statement I : The equivalent resistance of resistors in a series combination is smaller than least resistance used in the combination.

Statement II : The resistivity of the material is independent of temperature. In the light of the above statements, choose the correct answer from the options given below:

[2023 (15 Apr Shift 1)]

- (1) Both Statement I and Statement II are false
- (2) Both Statement I and Statement II are true
- (3) Statement I is true but Statement II is false
- (4) Statement I is false but Statement II is true

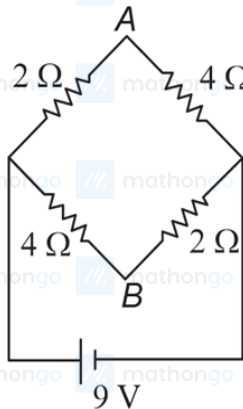
24. In the given circuit, the current I through the battery will be



[2023 (15 Apr Shift 1)]

- (1) 2.5 A
- (2) 1 A
- (3) 2 A
- (4) 1.5 A

25. A network of four resistances is connected to 9 V battery, as shown in figure. The magnitude of voltage difference between the points A and B is _____ V



[2023 (15 Apr Shift 1)]

ANSWER KEYS

1. (1) 2. (2) 3. (20) 4. (2) 5. (25) 6. (2) 7. (125) 8. (2)
9. (100) 10. (4) 11. (3) 12. (2) 13. (1) 14. (1) 15. (3) 16. (5)
17. (2) 18. (15) 19. (3) 20. (16) 21. (50) 22. (1) 23. (1) 24. (4)
25. (3)

1. (1)

A galvanometer is a tool used to either detect or quantify an electric current. The movement of a magnetic needle or a coil in a magnetic field, which is an essential component of a galvanometer, typically indicates the current and its strength.

Depending up on the resistance and the connectivity in the circuit, galvanometer can be used either as voltmeter or as an ammeter.

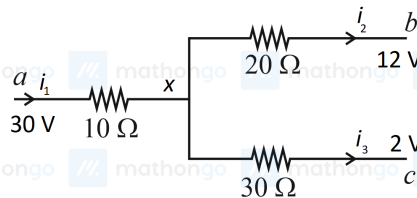
Voltmeter should have very high resistance, while ammeter should be having very less resistance.

Hence, the high resistor R_1 should be connected in series with the galvanometer in order to measure the potential difference across the test resistor.

Also, the low resistor R_2 should be connected in parallel with the galvanometer in order to measure the current across the test resistor.

Thus, this combination will work properly.

2. (2)



Let the potential be x V at the junction.

By using $V = IR$ we get,

$$30 - x = 10i_1$$

$$x - 12 = 20i_2$$

$$x - 2 = 30i_3$$

By using Kirchhoff's law,

$$i_1 = i_2 + i_3$$

$$\Rightarrow \frac{30-x}{10} + \frac{12-x}{20} + \frac{2-x}{30} = 0$$

$$\Rightarrow \frac{90-3x+18-1.5x+2-x}{30} = 0$$

$$\Rightarrow x = 20 \text{ V}$$

The current through the 20Ω resistor is

$$i_2 = \left(\frac{30-12}{20} \right) \text{ A}$$

$$\Rightarrow i_2 = \frac{8}{20} \text{ A} = 0.4 \text{ A}$$

3. (20)

Let the resistance of the voltmeter be $R \Omega$. The voltmeter is in parallel with the 5Ω . So the total resistance in the circuit is

$$\left(\frac{5R}{5+R} + 2 \right) \Omega$$

Applying Ohm's law in the whole circuit, $V = IR$,

$$I = \left(\frac{3}{2 + \frac{5R}{5+R}} \right) \text{ A}$$

It is given that the potential difference across 5Ω is 2 V .

Thus, by Ohm's law,

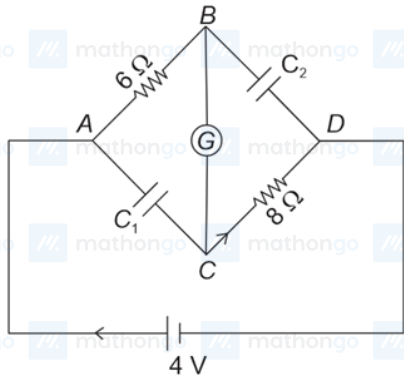
$$\left(\frac{3}{2 + \frac{5R}{5+R}} \right) \times \left(\frac{5R}{R+5} \right) = 2$$

$$\Rightarrow \frac{15+3R}{10+7R} \times \frac{5R}{5+R} = 2$$

$$\Rightarrow 15R = 20 + 14R$$

$$\Rightarrow R = 20 \Omega$$

4. (2)



Current flowing through both capacitors will be zero at steady state.

The potential difference across capacitor C_1 will be the sum of the potential difference across the resistor 6Ω and the galvanometer resistance.

Mathematically,

$$V_{C_1} = i[6\Omega + R_G] \dots (1)$$

Similarly, the potential difference across capacitor C_2 will be the sum of the potential difference across the resistor 8Ω and the galvanometer resistance. Mathematically,

$$V_{C_2} = i[R_G + 8\Omega] \dots (2)$$

Divide equation (1) by equation (2) to obtain the required ratio.

$$\begin{aligned} \frac{V_{C_1}}{V_{C_2}} &= \frac{i[6\Omega + R_G]}{i[R_G + 8\Omega]} \\ &= \frac{6\Omega + 2\Omega}{2\Omega + 8\Omega} \\ &= \frac{4}{5} \end{aligned}$$

5. (25)

The drift velocity is given by

$$v_d = \frac{I}{neA}$$

$$\Rightarrow I = nAev_d \dots (i)$$

The given data is

$$I = 2 \text{ A}$$

$$A = 25 \times 10^{-6} \text{ m}^2$$

$$n = 2 \times 10^{28}$$

$$e = 1.6 \times 10^{-19} \text{ C}$$

Substituting the given values in equation (i)

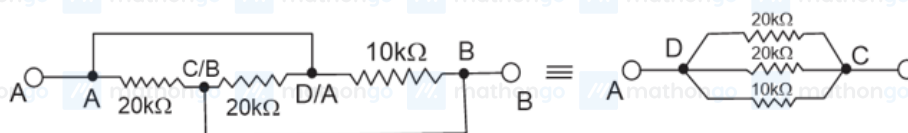
$$2 = 2 \times 10^{28} \times 25 \times 10^{-6} \times 1.6 \times 10^{-19} \times v_d$$

$$\Rightarrow v_d = \frac{2}{2 \times 10^{28} \times 25 \times 10^{-6} \times 1.6 \times 10^{-19}}$$

$$\Rightarrow v_d = \frac{1}{25 \times 1.6 \times 1000} \text{ m s}^{-1} = 25 \times 10^{-6} \text{ m s}^{-1}$$

6. (2)

The equivalent circuit of the given configuration can be drawn as follows:



With reference to the above diagram, the equivalent resistance of the given configuration can be calculated as follows:

$$\begin{aligned} \frac{1}{R} &= \frac{1}{20\Omega} + \frac{1}{20\Omega} + \frac{1}{10\Omega} \\ &= \frac{1}{5\Omega} \\ \Rightarrow R &= 5\Omega \end{aligned}$$

7. (125)

The formula to calculate the current through any conductor is given by

$$I = nAev_d \dots (1)$$

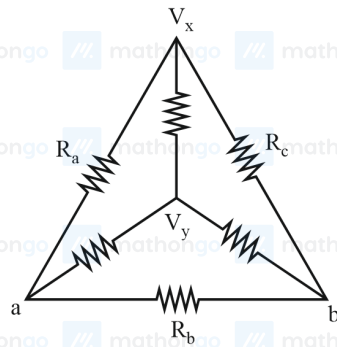
Substitute the values of the given parameters into equation (1) and solve to calculate the required drift velocity of the electrons.

$$3.2 = 8 \times 10^{28} \times 2 \times 10^{-6} \times 1.6 \times 10^{-19} \times v_d$$

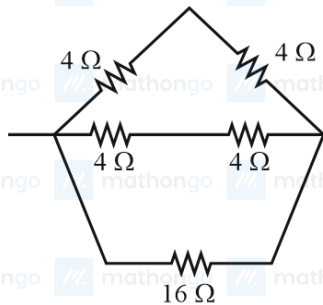
$$\Rightarrow v_d = \frac{3.2}{16 \times 1.6 \times 1000} \text{ m s}^{-1}$$

$$= 125 \times 10^{-6} \text{ m s}^{-1}$$

8. (2)



The resistance in ab is parallel to all other resistances. Since all the resistances other than R_b is 4Ω the potential is zero ($V_x = V_y$) as the upper branch will become a case of balanced wheatstone bridge between V_x & V_y . So, the simplified circuit looks like the one below.



Hence, the equivalent resistance is

$$\frac{1}{R_{eq}} = \frac{1}{8} + \frac{1}{8} + \frac{1}{16} = \frac{5}{16}$$

$$\Rightarrow R_{eq} = \frac{16}{5} \Omega = 3.2 \Omega$$

9. (100)

When all the resistors are connected in series, the total resistance is

$$R_{max} = 10R$$

$$\Rightarrow R_{max} = 10 \times 10 = 100 \Omega$$

When all the resistors are connected in parallel, the total resistance is

$$R_{min} = \frac{R}{10}$$

$$\Rightarrow R_{min} = \frac{10}{10} = 1 \Omega$$

Hence, the ratio,

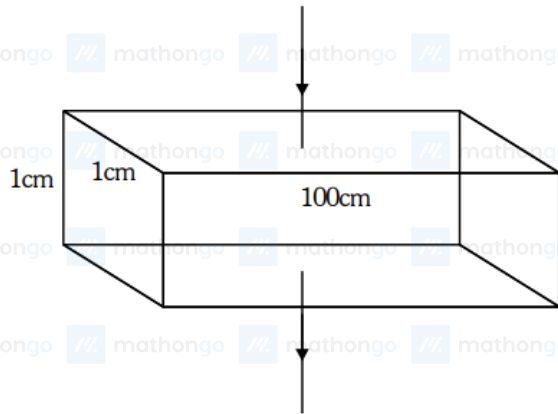
$$\frac{R_{max}}{R_{min}} = \frac{100 \Omega}{1 \Omega} = 100.$$

10. (4)

The electrons inside the conductor move with an average velocity due to the applied field in a direction opposite to the field, this velocity is known as drift velocity. Electric field are applied from high to low potential region of the conductor. Also, electrons are negatively charged.

Electron drifts from lower potential to higher potential on curved path.

11. (3)



It is given that

$$l = 1 \text{ cm} = 0.01 \text{ m}$$

$$A = 100 \text{ cm} \times 1 \text{ cm} = 100 \text{ cm}^2$$

$$\Rightarrow A = 0.01 \text{ m}^2$$

$$\rho = 3 \times 10^{-7} \Omega \text{ m}$$

Using the formula of resistance,

$$R = \frac{\rho l}{A} = \frac{3 \times 10^{-7} \times 0.01}{0.01} = 3 \times 10^{-7} \Omega$$

12. (2)

Let the resistance of the filaments be R and the voltage be V .

For the series combination the power is

$$P_1 = \left(\frac{V^2}{2R} \right)$$

For parallel combination,

$$P_2 = \frac{V^2}{\left(\frac{R}{2} \right)} = \left(\frac{2V^2}{R} \right)$$

Hence, the ratio is

$$\Rightarrow \frac{P_2}{P_1} = \frac{4}{1}$$

13. (1)

The current sensitivity is given as

$$S_i = \frac{NBA}{C}$$

The voltage sensitivity is

$$S_v = \left(\frac{NAB}{CR} \right), \text{ where } R \text{ is the galvanometer resistance.}$$

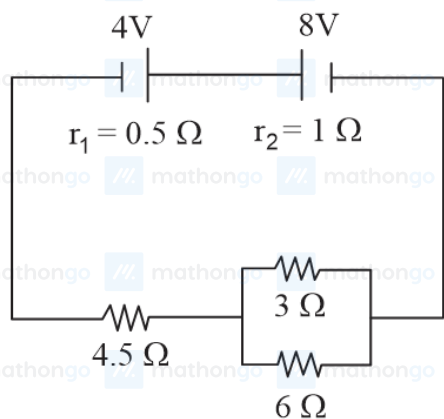
Since the resistance is constant,

$$\Delta S_i = (\Delta S_v)$$

Hence, the percentage change in voltage sensitivity is,

$$(\Delta S_v) = +25\%$$

14. (1)

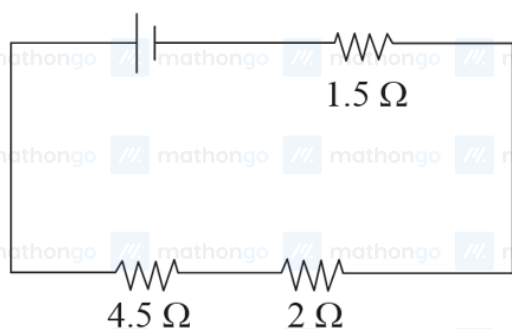


$$E_2 - E_1 = 8 - 4 = 4V$$

Let the current flowing through the circuit be I A in anti-clockwise direction.

The net resistance is $R = 4.5 \Omega + 2 \Omega + 1.5 \Omega = 8 \Omega$

Hence, $I = \frac{V}{R} = \frac{4V}{8\Omega} = 0.5$ A

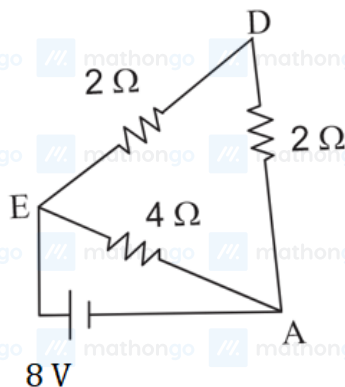
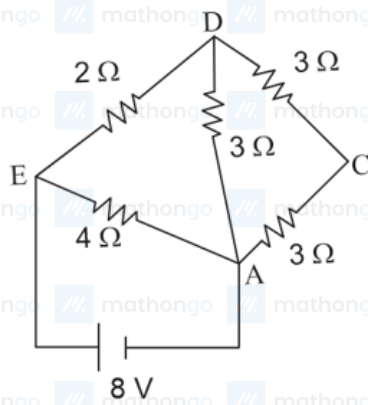
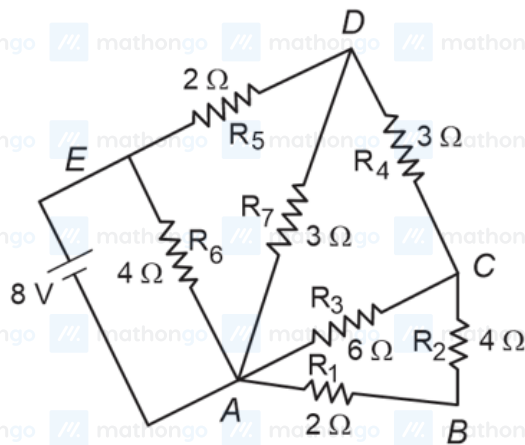


The current flowing through the 3 Ω resistor is

$$\frac{6}{6+3} \times 0.5 = \frac{1}{3} \text{ A}$$

So, $x = 1$

15. (3)



As can be seen from the first circuit above, CB & AB are in series which is in parallel with CA . So, the net resistance is $R' = \frac{6}{2} \Omega = 3 \Omega$ as shown in the second diagram.

From the second diagram, DC & CA is in series which is parallel to DA . Hence the resistance is

$$R'' = \left(\frac{6 \times 3}{6+3} \right) \Omega = 2 \Omega \text{ as shown in third diagram.}$$

From third diagram, both ED & DA are in series which is parallel to EA .

The equivalent resistance as can be seen from the above simplified circuit is $R_{eq} = 2 \Omega$.

The current I through battery, $I = \frac{8}{2} = 4 \text{ A}$.

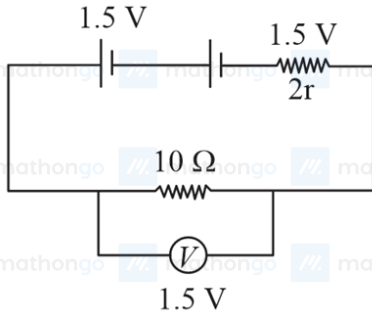
From the second diagram, the part $EDACD$ has a resistance of 4Ω , the total current will divide equally between $EDACD$ & EA . So consider $I_1 = 2 \text{ A}$ flowing through ED . Note that I_1 will be the same current flowing in the first diagram through ED .

From first diagram, when the resistances are in parallel the current is divided in the inverse ratio. Thus,

$$I' \text{ through } CD = \frac{3}{9} \times 2 = \frac{2}{3} \text{ A}$$

$$I'' \text{ through } R_2 = \frac{1}{2} \left(\frac{2}{3} \right) = \frac{1}{3} \text{ A}$$

16. (5)



Let $R_1 = 10 \Omega$.

Using $V = iR$

$$i = \left(\frac{3}{10+2r} \right)$$

The voltmeter reading is $V' = 1.5 \text{ V}$.

Thus,

$$iR_1 = V'$$

$$\Rightarrow 10 \times \frac{3}{10+2r} = 1.5$$

$$\Rightarrow 20 = 10 + 2r$$

$$\Rightarrow 2r = 10$$

$$\Rightarrow r = 5 \Omega$$

17. (2)

The formula to calculate the resistance of the original wire is given by

$$R = \rho \frac{l}{A} \dots (1)$$

When the new wire is formed, its resistance is given by

$$R' = \rho \frac{l}{4A} \dots (2)$$

Divide equation (2) by equation (1) and simplify to obtain the expression for the new resistance.

$$\frac{R'}{R} = \frac{\rho \frac{l}{4A}}{\rho \frac{l}{A}}$$

$$\Rightarrow R' = R \frac{l}{4l} \dots (3)$$

Substitute the values of the known parameters into equation (3) to calculate the required new resistance.

$$R' = \frac{l}{4l} \times 160 \Omega$$

$$= 10 \Omega$$

18. (15)

By Ohm's law, $V = IR$.

Let the current and resistance at 0°C be i_0 & R_0 respectively.

Let the current and resistance at 100°C be i_{100} & R_{100} respectively.

By the given condition,

$$i_0 R_0 = i_{100} R_{100}$$

$$\Rightarrow 2R_0 = 1.2R_0(1 + 100\alpha)$$

$$\Rightarrow 1 + 100\alpha = \frac{5}{3}$$

$$\Rightarrow 50\alpha = \frac{1}{3}$$

Since the voltage remains constant,

$$i_{50} R_{50} = i_0 R_0$$

$$\Rightarrow i_{50} = \frac{i_0 R_0}{R_{50}} = \frac{2 \times R_0}{R_0(1 + 50\alpha)} = \frac{2}{1 + \frac{1}{3}} = 1.5 \text{ A}$$

$$\Rightarrow i_{50} = 15 \times 10^2 \text{ mA}$$

19. (3)

The resistances can be calculated as

$$R_A = R + \frac{R}{2} = \frac{3R}{2}$$

For second circuit

$$\frac{1}{R_B} = \frac{1}{2R} + \frac{1}{R} = \frac{3}{2R}$$

$$\Rightarrow R_B = \frac{2R}{3}$$

For third circuit

$$R_C = \frac{R}{3}$$

For fourth circuit

$$R_D = 3R$$

The formula for power is given by $P = I^2 R$.

Clearly,

$$P_D > P_A > P_B > P_C$$

20. (16)

The formula to calculate the power dissipated in the first case can be written as

$$P_1 = \frac{V_0^2}{R} \dots (1)$$

When the wire is halved, its resistance will also become half the previous value.

Thus, the formula to calculate the power dissipated in the second case can be written as

$$P_2 = \frac{V_0^2}{\frac{R}{2}} \times 2$$

$$= \frac{4V_0^2}{R} \dots (2)$$

Divide equation (2) by equation (1) to calculate the required ratio.

$$\frac{P_2}{P_1} = \frac{\frac{4V_0^2}{R}}{\frac{V_0^2}{R}}$$

$$= 4$$

$$= \sqrt{16}$$

Hence, $x = 16$.

21. (50)

The formula to calculate the maximum current through the galvanometer, when a shunt resistance r_1 is connected is given by

$$(i_G)_{\max} = \frac{r_1}{r_1 + R_G} i_{\max} \dots (1)$$

When a series resistance r_2 is connected, the potential difference across the galvanometer is given by

$$V = (i_G)_{\max} [R_G + r_2] \dots (2)$$

From equations (1) and (2), it can be written that

$$V = \frac{r_1}{r_1 + R_G} i_{\max} (R_G + r_2)$$

$$\Rightarrow V r_1 + V R_G = i_{\max} r_1 R_G + i_{\max} r_1 r_2$$

$$\Rightarrow (V - i_{\max} r_1) R_G = i_{\max} r_1 r_2 - V r_1$$

$$\Rightarrow R_G = \frac{i_{\max} r_1 r_2 - V r_1}{V - i_{\max} r_1} \dots (3)$$

Substitute the values of the known parameters into equation (3) to calculate the required galvanometer resistance.

$$R_G = \frac{0.250 \text{ A} \times 5 \Omega \times 1050 \Omega - 25 \text{ V} \times 5 \Omega}{25 \text{ V} - 0.250 \text{ A} \times 5 \Omega}$$

$$= 50 \Omega$$

22. (1)

The resistance is given by

$$R = \frac{V}{I_g} - G$$

$$\Rightarrow \frac{50}{10^{-3}} - 54 \approx 50 \text{ k}\Omega$$

For ammeter,

$$S = \frac{I_g G}{I - I_g}$$

$$\Rightarrow S = \frac{10^{-3} \times 54}{(10 - 1) \times 10^{-3}} = 6 \Omega$$

23. (1)

The formula for series equivalent resistance is given by

$$R_s = R_1 + R_2 + \dots + R_n$$

Hence, the equivalent resistance will be the highest value compared to other resistances.

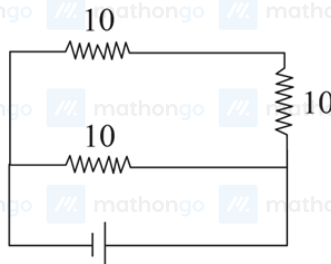
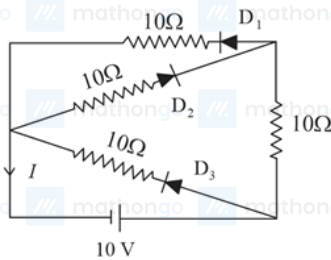
The resistivity of a material is given by,

$$\rho = \rho^0(1 + \alpha \Delta T)$$

Clearly, the resistivity is dependant on temperature.

Hence, this is the right option.

24. (4)



In the first circuit diagram, the diodes D_1 & D_3 are forward biased and D_2 is reverse biased (Hence, no current through it). So the circuit is simplified to the second diagram.

Hence, the equivalent resistance is

$$R_{eq} = \frac{20 \times 10}{20 + 10}$$

$$\Rightarrow R_{eq} = \frac{20}{3} \Omega$$

Using Ohm's law, the current is

$$I = \frac{10}{\frac{20}{3}} = \frac{30}{20} \text{ A} = 1.5 \text{ A}$$

25. (3)

The resistance in each branch is 6Ω . They are in parallel. So, net resistance is $R' = \frac{6}{2} = 3 \Omega$.

The total current flowing through the loop,

$$I = \frac{V}{R} = \frac{9 \text{ V}}{3 \Omega} = 3 \text{ A}$$

Thus, through each branch the current is 1.5 A .

Hence, the voltage difference across the points is

$$V_A - V_B = (4 \times 1.5) - (2 \times 1.5) = 3 \text{ V}$$